Pinhole Aperture Optics for Myopia Correction

Theory, Simulation, and Photographic Validation of a Reduced-Eye Diffraction/Defocus Model

IYPT Team

2026

Presentation Roadmap

Part 1 (this deck)

Introduction → Theoretical Model → Evaluators/Metrics Overview

Later parts (not in this deck)

Experiment 1 (ExP2 photographic validation) → Experiment 2 (ExP3 optical bench) → Array design & fabrication → Discussion

How this deck is organized

Every slide from the main talk is followed, where relevant, by **RESERVE** slides carrying the full derivation, code, or extended data. Nothing is cut – reserve slides exist for defending any question in depth during the jury discussion.

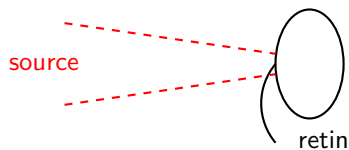
Section 1: Introduction

1 Introduction

2 Theoretical Model

The Problem: Correcting Myopia Without a Lens

- A myopic (short-sighted) eye focuses distant light **in front of** the retina, not on it – the retinal image is a blur disc, not a point.
- Standard correction: a diverging lens (glasses/contacts) restores focus.
- **Pinhole optics** offer a lens-free alternative: a small aperture placed in front of the eye restricts the cone of rays reaching the retina, shrinking the blur disc geometrically – *without* refracting light at all.
- This is the physical basis of pinhole glasses, marketed as a correction-free vision aid.



A myopic eye: the marginal rays (dashed) converge in front of the retina, spreading into a blur circle at the retinal plane.

Why This Is a Genuine Physics Problem, Not Just Engineering

- Shrinking the aperture reduces **geometric** blur linearly with aperture diameter – naively, smaller is always better.
- But a small aperture increases **diffraction** blur (Airy-disk scale $\propto 1/a_{\text{hole}}$) – naively, larger is always better.
- These two effects trade off oppositely with aperture size $\Rightarrow$ there is an **optimum aperture** for every myopic power M, not a "smaller is always better" answer.
- This project's core question: *find that optimum, validate it against real photographic measurement, and check it against independent clinical literature.*

This is exactly the geometric/diffraction competition this project's full theoretical model (Section 2) is built to resolve quantitatively.

Project Scope – What Was Actually Built and Tested

Theory

- Full wave-optics (Angular Spectrum Method) simulation
- Closed-form RSS approximation, cross-checked against it
- Two-pinhole ghost-image separation formula (validated, $R^2 = 0.995$)
- Gaze-weighted safety/comfort evaluation

Experiment

- ExP2: real point-source photography, 5 apertures, phone camera
- ExP3: independent frosted-screen optical-bench sweep, 6 apertures

Cross-checks

- Clinical literature anchor (Miller & Johnson 1977, via Shen 2026)
- No-reference image-quality scoring (BRISQUE) on real scene photos

Design output

- Gaze-weighted multi-pinhole array design, per myopic power M
- A real photochemical-etching fabrication specification (0.05mm brass shim, UV photoresist)

This deck's Part 1 covers Introduction + Theoretical Model only.

Research Questions

1. What is the theoretically optimal pinhole aperture at each myopic power M, once diffraction is modeled properly (not just geometric optics)?
2. Does a real photographic measurement of blur growth with M match that theory – and if not, where and why?
3. Does a simpler closed-form approximation (RSS) do just as well as the full wave-optics simulation, or is the extra complexity justified?
4. Does this project's own theoretical aperture-tolerance prediction agree with independent clinical literature?
5. Can a *multi-pinhole array* (rather than one hole) be designed that stays ghost-free across natural gaze movement, and can it actually be fabricated?

Section 2: Theoretical Model

1 Introduction

2 Theoretical Model

2.1 The Reduced-Eye Model

- Every theoretical prediction in this project uses a single-surface **reduced eye**: a thin lens of fixed power P_0, immersed on its image side in vitreous of refractive index n'.
- Myopic defocus is modeled as an **added power** M (diopters) at the eye/lens – exactly as an induced-refractive-error lens would.
- Image-side (media-reduced) focal length:

$$f'_{\text{image}} = \frac{n'}{P_0}$$

Not the naive $f' = 1/P_0$

That simpler form is only correct if image space were air. This project's own source code uses `refractive_index_correction=True` throughout, so n' must appear.

Constants used everywhere in this project (traced to source, not assumed):

$$P_0 = 60.0 \text{ D}, \qquad n' = 1.336, \qquad \lambda_{\text{vacuum}} = 550 \text{ nm}$$

$$\Rightarrow f'_{\text{image}} = \frac{1.336}{60.0} \times 10^6 \ \mu\text{m} = 22266.667 \ \mu\text{m}$$

2.2 The Blur Metric: Encircled-Energy Radius r_{EE50}

- Every model and every measurement in this project reduces to one number per (aperture, defocus) condition: the radius that encloses 50% of the total collected optical energy.
- Chosen because it is well-defined for *both* a simulated point-spread function *and* a real, noisy photograph of a point source – unlike a fitted Gaussian σ, which assumes a shape the real data may not follow.

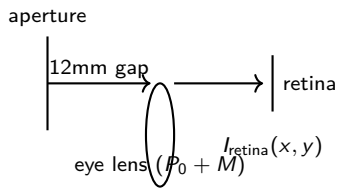
$$\sum_{r_i \le r_{EE50}} I(r_i) \;=\; 0.5 \times \sum_{i} I(r_i)$$

Used identically in three places

- ASM simulation table (physical μm, from the retinal irradiance map)
- ExP2 photographic pipeline (pixels, background-subtracted blob)
- RSS closed-form estimate (physical μm, analytic)

2.3 Model A – Full Wave-Optics Simulation (ASM)

- The primary theoretical prediction is **not** an analytic formula – it is a full scalar-diffraction simulation: the **Angular Spectrum Method**.
- Propagates a coherent field from the aperture, through a 12mm air gap, through the eye lens (adding power $P_0 + M$ as a quadratic phase), to the retina.
- Run once, from scratch, across a grid of 15 aperture sizes $\times$ 8 myopic powers M (120 points) $\rightarrow$ the single theory table (`blur_sweep.csv`) every comparison in this project uses.



Grid: 4096×4096, 6mm window (dx $\approx 1.465\,\mu$m/px) – this exact pixel size later sets the measurement-uncertainty floor used in the ghost-margin safety framework.

2.4 Model B – Closed-Form Approximation (RSS)

- ASM is expensive (a full 4096×4096 propagation per point) and opaque as a design tool. RSS combines geometric and diffraction blur in quadrature – a much faster, simpler estimator, checked independently against the same measured photographs ASM is checked against.

$$c_{\text{geom}}(a, M) = \frac{a \cdot M}{P_0} \qquad c_{\text{diff}}(a) = 1.22\,\frac{\lambda_{\text{eff}}\, f'}{a}$$

$$c_{\text{RSS}}(a, M) = \sqrt{c_{\text{geom}}(a, M)^2 + c_{\text{diff}}(a)^2}$$

Headline result (full data in Results deck, not here)

ASM matches the measured photographs decisively better than RSS across every comparable case (18/18 rows). RSS is kept because it is the *only* model computable with no measured/simulated $M = 0$ reference – useful, but not the primary predictive model.

2.5 Model C – Two-Pinhole Ghost-Image Separation

- For a multi-pinhole **array** design: two neighboring holes each form their own retinal image of a point source. Under defocus, these separate spatially – a visible "ghost" once the separation exceeds the eye's own resolvable blur.

$$\Delta r(s, M) = \frac{s \cdot M}{P_0 - M}$$

Independently validated

$R^2 = 0.995$ against a separate two-hole coherent wave-optics simulation (32 points), vs. $R^2 = -1701.71$ for an earlier, disproven formula.

This formula, combined with a safety-margin framework (Section 2.7), is what makes a physically buildable *array* design possible, not just a single optimal hole.

2.6 Evaluators / Metrics – What This Project Actually Measures

Evaluator	What it quantifies
r_{EE50}	Blur radius – primary metric, theory *and* both experiments
RSS vs. ASM % deviation	Which theoretical model actually fits measured data
Ghost margin (worst-case)	Hard safety gate: is a design ghost-free at *every* tested gaze angle
Gaze-weighted ghost/blind %	Realistic-use metric: % of actual viewing *time* spent ghosting/blind, using measured eye-tracking eccentricity data
LogMAR / acuity-optimal aperture	Clinical-acuity-based optimum, cross-checked against the pure-blur optimum
BRISQUE (no-reference IQA)	Real-scene photo quality score, independent of the point-source pipeline

Why so many evaluators, not just one

Worst-case and gaze-weighted answer *different* questions (hard guarantee vs. realistic experience) – this project reports both, deliberately, rather than picking one and hiding the other.

2.7 Safety-Margin Framework (Why a Margin, Not Just a Sign)

- A design "passes" only if its margin clears a **known, sourced uncertainty** in the underlying measurement – not merely margin > 0.

$$\text{margin} = 1 - \frac{\Delta r_{\text{actual}}}{r_{\text{resolvable}}}, \qquad \text{threshold} = \frac{2.0 \times \delta_{r,\text{unc}}}{r_{\text{resolvable}}}$$

$$\delta_{r,\text{unc}} = 2 \times dx_{\text{grid}} = 2 \times 1.465\,\mu\text{m} = 2.93\,\mu\text{m}$$

Audit history, reported honestly (full trail in Reserve)

An earlier version of this framework required only margin > 0 – found to silently reward a design with *zero illuminated holes* (infinite margin, zero usefulness). Fixed by adding an independent viability gate.

Section 2 Summary – What Carries Into the Rest of the Talk

- One consistent eye model ($P_0 = 60$D, $n' = 1.336$) feeds every theoretical number in this project – no separately invented constants.
- Two independent theoretical models (ASM, RSS) exist so the photographic experiments (Part 2) test *model choice*, not just *model fit*.
- The ghost-separation formula and safety-margin framework (Section 2.5/2.7) are what let this project go from "one optimal hole" to "a real, checkable multi-pinhole array design" (Part 3).
- Every evaluator above will reappear with real numbers in the Results parts of this talk.

Next: Experiment 1 – Photographic Validation (Part 2, separate deck)

RESERVE SLIDES

Full derivations, source code, and extended data for every model and evaluator in Section 2.

Not part of the timed talk – called up during jury Q&A as needed.

R1 – ASM: Full Simulation Pipeline

RESERVE

1. Circular aperture radius a_{hole} illuminated by a coherent plane wave, $\lambda = 550\text{nm}$ vacuum.
2. Angular-spectrum free-space propagation, 12mm gap, on a 4096×4096 grid spanning a 6mm window ($dx = 6000/4096 \approx 1.465\,\mu\text{m/pixel}$).
3. Eye/lens applies a quadratic phase for total power ($P_0 + M$), using the reduced, media-corrected convention.
4. Propagate the eye's own image-side focal distance f' to the retina $\Rightarrow$ irradiance map $I_{\text{retina}}(x, y)$.
5. $r_{EE50/70/80/90}$ and RMS spot radius computed by radial integration of I_{retina}.

Grid history (reported honestly): an earlier run used a 2048-point grid; found to give r_{EE50} values exactly $2\times$ too large for large apertures (a known grid-resolution artifact). Fixed to 4096; the stale file was preserved, not deleted (`blur_sweep_gridN2048_legacy.csv`).

R1b – ASM Sweep: Source Code

RESERVE

```
result = p2.run_pipeline_point(
    a_hole=a_um * 1e-6, wavelength=WAVELENGTH_NM * 1e-9,
    d_gap=D_GAP_MM * 1e-3, grid_N=GRID_N, window_L_mm=WINDOW_L_MM,
    M_diopters=M, P0=60.0, n_prime=1.336,
    D_pupil_um=D_PUPIL_UM, refractive_index_correction=True,
)
if result["junction_check_passed"]:
    I_retina = result["I_retina"]
    dx = result["cornea_result"]["dx"]
    r_edges, EE, total_power = junction.encircled_energy(I_retina, R, dx)
    for p in PERCENTILES:
        row[f"r_EE_{int(p*100)}_um"] = junction.r_for_EE(r_edges, EE, p) * 1e6
```

new_sweeps/run_blur_sweep.py – generates blur_sweep.csv, the single theory source table for this entire project.

R2 – RSS: Full Derivation

Geometric defocus term – a defocus of M diopters, with total eye power P_0, produces a ray-optics blur circle scaling linearly with aperture radius and M relative to P_0:

$$c_{\text{geom}}(a, M) = \frac{a \cdot M}{P_0}$$

Diffraction term – Airy-disk-scale blur at the retina, using the *effective (media-reduced)* wavelength, not the raw vacuum value:

$$c_{\text{diff}}(a) = 1.22\,\frac{\lambda_{\text{eff}}\, f'}{a}, \qquad \lambda_{\text{eff}} = \frac{\lambda_{\text{vacuum}}}{n'} = \frac{0.550}{1.336}\,\mu\text{m} = 0.4117\,\mu\text{m}$$

Why the media-reduced wavelength

Pairing the media-reduced $f' = n'/P_0$ with the raw vacuum wavelength would under-apply the lens's converging phase by a factor of n' – this project's own source module explicitly warns against exactly that inconsistency.

Normalization: at $M = 0$, $c_{\text{geom}} = 0$ identically, so $c_{\text{RSS}}(a, 0) = c_{\text{diff}}(a)$ alone – the only model needing no measured/simulated $M = 0$ reference at all.

R2b – RSS: Source Code

RESERVE

```
P0 = 60.0
N_PRIME = 1.336
LAMBDA_VACUUM_UM = 550.0 / 1000.0
LAMBDA_UM = LAMBDA_VACUUM_UM / N_PRIME # media-reduced
F_PRIME_UM = (N_PRIME / P0) * 1e6

def c_geom_um(a_hole_um, M):
    return a_hole_um * M / P0

def c_diff_um(a_hole_um):
    return 1.22 * LAMBDA_UM * F_PRIME_UM / a_hole_um

def c_tot_rss_um(a_hole_um, M):
    return math.sqrt(c_geom_um(a_hole_um, M)**2 + c_diff_um(a_hole_um)**2)

def predicted_ratio_rss(a_hole_um, M):
    return c_tot_rss_um(a_hole_um, M) / c_tot_rss_um(a_hole_um, 0.0)
```

ExperimentalComparison/rss_comparison.py

R3 – Ghost-Separation Formula: Derivation & Validation

RESERVE

$$\Delta r(s, M) = \frac{s \cdot M}{P_0 - M}$$

- s = pitch (center-to-center hole separation), M = myopic power, P_0 = eye power.
- Derived from the reduced-eye axial model (myopia_axial.py) – the v_r/v_f axial form and this M-only form are algebraically identical under that model; both are kept in the source module so the identity can be *checked*, not assumed.

Validation history:

- An earlier "legacy" formula was tested against an independent two-hole coherent wave-optics propagation simulation (32 points): $R^2 = -1701.71$ – badly wrong, disproven and retired.
- The corrected form above: $R^2 = 0.995135$ against the same independent simulation.

R3b – Ghost Formula: Source Code

RESERVE

```
def delta_r_corrected_um(pitch_um, v_r_um, v_f_um):
    """Eq. (C): delta_r = s * |v_r - v_f| / v_f. Vanishes at v_r = v_f (M=0)."""
    return pitch_um * abs(v_r_um - v_f_um) / v_f_um

def delta_r_corrected_from_M_um(pitch_um, M_diopters, P0=60.0):
    """Eq. (C'): the same quantity written in terms of myopia only,
    s * M/(P0-M). Identical to delta_r_corrected_um() under the axial model."""
    return pitch_um * M_diopters / (P0 - M_diopters)
```

array_calculation/engine/ghost_formula.py – validated against stage0_ghost_validation/validate_ghost_formula.py.

R4 – Safety-Margin Framework: Full Derivation

RESERVE

$$\text{margin} = 1 - \frac{\Delta r_{\text{actual}}}{r_{\text{resolvable}}}$$

$$\delta_{r,\text{unc}} = 2 \times dx_{\text{grid}} = 2 \times 1.46484375\,\mu\text{m} = 2.9296875\,\mu\text{m}$$

$$\text{threshold} = \underbrace{2.0}_{\text{safety multiple}} \times \frac{\delta_{r,\text{unc}}}{r_{\text{resolvable}}}$$

- dx_{grid} is the ASM simulation's own grid pixel size (R1).
- Factor of 2 in $\delta_{r,\text{unc}}$: the same source module's own stated tolerance for a two-*peak* separation measurement (± 1 pixel per peak).
- Safety multiple 2.0: this project's own established convention, order-matched to the $2\times$ already used for the pixel tolerance.

Audit fix, reported honestly: the original two gates (`gate_ghost_safety`, `gate_vignetting_safety`) only tested for *absence* of ghosting – a design with **zero illuminated holes** has zero active pairs, hence infinite margin on both, hence scored as maximally safe. A third, independent `gate_viability` was added (illuminated area $\geq 20\%$ of single-hole area) to close this without folding it into the other two gates.

R4b – Gates: Source Code

RESERVE

```
GHOST_FORMULA_GRID_DX_UM = 1.46484375
GHOST_DELTA_R_UNCERTAINTY_UM = 2.0 * GHOST_FORMULA_GRID_DX_UM
MARGIN_SAFETY_MULTIPLE = 2.0
VIABILITY_MIN_FRACTION_OF_HOLE_AREA = 0.20

def ghost_margin_threshold(df):
    return MARGIN_SAFETY_MULTIPLE * GHOST_DELTA_R_UNCERTAINTY_UM / df.resolvable_blur_um

def gate_ghost_safety(df):
    return df.worst_case_ghost_margin >= ghost_margin_threshold(df)

def gate_viability(df):
    """Requires␣illuminated_area␣>=␣0.20*pi*R_h^2␣at␣EVERY␣tested␣gaze␣angle,
␣␣␣␣not␣merely␣at␣phi=0␣--␣closes␣the␣'rewards␣blindness'␣degenerate␣case."""
    ...
```

array_evaluation/comparison_metric/gates.py

R5 – Gaze-Weighted Evaluation: Full Math

RESERVE

$$P(\text{blind}) = \int w(\varphi) \cdot \mathbb{1}[\text{blind at } \varphi]\, d\varphi, \qquad P(\text{ghost}) = \int w(\varphi) \cdot \mathbb{1}[\text{ghosting at } \varphi]\, d\varphi$$

where $w(\varphi)$ is a **measured** eye-tracking eccentricity distribution, normalized to sum to 1:

Eccentricity	0–1°	5–6°	10–11°	16–17°	>25°
% of viewing time	9.29%	7.57%	4.40%	1.45%	0.58%

- Weight applies to the binary *outcome*, not the raw margin – averaging raw margins can hide a rare-but-total failure inside a good mean.
- Every angle sampled across *all azimuths*, not one radial direction – ring-based geometries were found to blind out specifically in the *gaps between* discrete ring holes.
- Deliberately additive: does not replace the worst-case gate (R4) as the primary safety criterion.

R6 – BRISQUE: What It Actually Measures

RESERVE

- **No-reference** image quality metric – needs no "ground truth" sharp image to compare against, unlike PSNR/SSIM.
- Trained on natural-scene statistics: fits a generalized Gaussian distribution to local normalized pixel-intensity statistics and scores how far an image's statistics deviate from natural, undistorted images.
- Lower score = closer to natural/undistorted; higher = more perceived distortion (blur, noise, compression artifacts all raise it).
- Used here on ExP1's real *scene* photographs (not a point source) – a natural-image metric is appropriate precisely because these are natural scenes, unlike the point-source r_{EE50} pipeline.

```
cv2.quality.QualityBRISQUE_compute(img, model.yml, range.yml)
```

OpenCV's BRISQUE model files (`brisque_model_live.yml`, `brisque_range_live.yml`) are the standard pretrained SVM regressor from the LIVE image-quality database – not retrained on this project's own data.

R7 – LogMAR / Acuity-Optimal Aperture

RESERVE

- A second, independent optimum: instead of minimizing pure blur radius, minimize **predicted logMAR acuity** (a clinical acuity scale) at each M.
- Computed on the same 15×8 (aperture $\times$ M) grid as the ASM blur sweep, so the two optima are directly comparable.

M (D)	blur-optimal (µm)	acuity-optimal (µm)	logMAR	gap
0	700	2000	-0.344	185.7%
2	750	750	0.076	0.0%
4	500	600	0.181	20.0%
8	400	400	0.331	0.0%

The two optima *agree exactly* at some M (e.g. $M = 2, 8$) and diverge sharply at others (e.g. $M = 0$: 185.7% gap) – a real, reported finding, not smoothed to a single "the optimum is X" number.

RESERVE

R8 – Full Constants Table (Every Number, Traced to Source)

Constant	Value	Source
P_0 (eye power)	60.0 D	`run_blur_sweep.py` l.56
n' (vitreous index)	1.336	`run_blur_sweep.py` l.56
λ_{vacuum}	550 nm	`run_blur_sweep.py` l.32
Air gap (aperture→eye)	12 mm	`run_blur_sweep.py` l.30
ASM grid	4096^2, 6mm window	`run_blur_sweep.py` l.32
dx_{grid}	1.46484375 µm	derived: 6000/4096
$\delta_{r,\text{unc}}$	2.9296875 µm	`gates.py` l.89
Margin safety multiple	2.0×	`gates.py` l.91
Ghost formula validation	$R^2 = 0.995135$	`validate_ghost_formula.py`

Every number on this slide was located in this project's own source files during this analysis, not assumed or independently invented.